

Clean Energy **Awareness** Assistant

Your AI Mentor for Sustainable Energy Solutions

1M1B AI for Sustainability Virtual Internship

Aligned with UN SDG 7: Affordable & Clean Energy

The Core Challenge

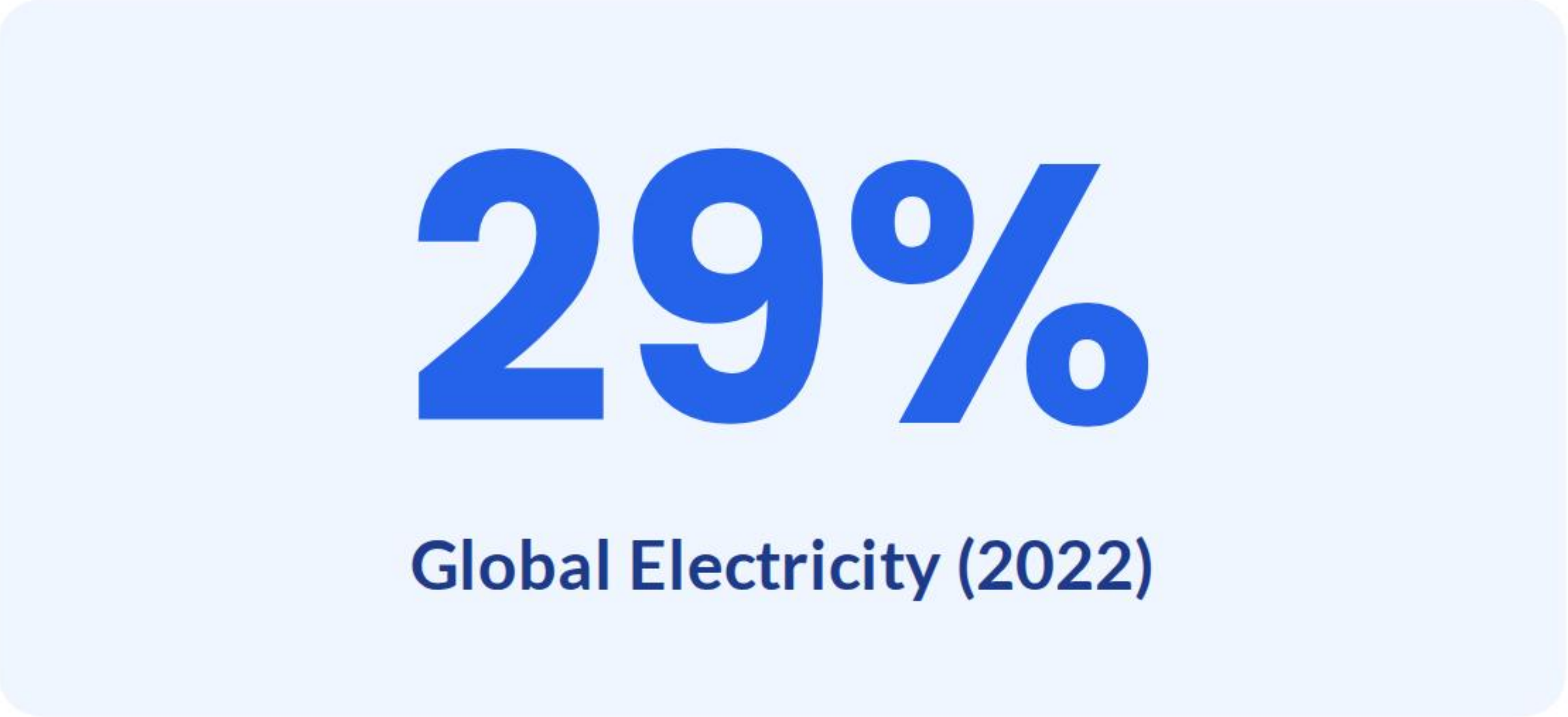
Addressing the global energy awareness gap and combating misinformation around sustainable infrastructure implementation among students and institutions.

The Energy Knowledge Gap

- **Technical Complexity:** Understanding the transition to renewable sources like solar or wind is often overly technical for non-engineering students.
- **Lack of Guidance:** Students and campus administrators struggle to find actionable, zero-cost steps to adopt energy efficiency.
- **Financial Misinformation:** Many believe green transitions are financially impossible, unaware that solar costs have dropped over 90% since 2010.
- **Urgency of SDG 7:** Achieving universal access to modern energy requires immediate, grass-roots awareness.



Driving Towards UN SDG 7



29%

Global Electricity (2022)

Accelerating Renewable Adoption

As integrated in our AI Assistant's knowledge base, renewable energy accounted for **29% of global electricity generation** in 2022.

Transitioning to clean energy is not just an environmental imperative; every dollar invested in renewable energy creates **3 times more jobs** than fossil fuels, making it a critical economic driver for the future.

The Solution: An AI Mentor

Powered by IBM BOB

We developed a lightweight, single-page web application leveraging IBM BOB's generative capabilities, ensuring rapid deployment without complex coding.

Targeted Prompt Engineering

The AI is strictly constrained via system prompts to act as a 'Clean Energy Awareness Mentor', providing facts, statistics, and actionable advice aligned with UN SDG 7.

Accessible User Interface

Designed with a clean blue and yellow theme, the app offers an intuitive chat interface making sustainable education engaging for everyone.



Core Application Features



Interactive Chat

Users can ask direct questions about solar viability, wind power, or energy efficiency, receiving detailed, fact-based guidance instantly.



Quick Green Quiz

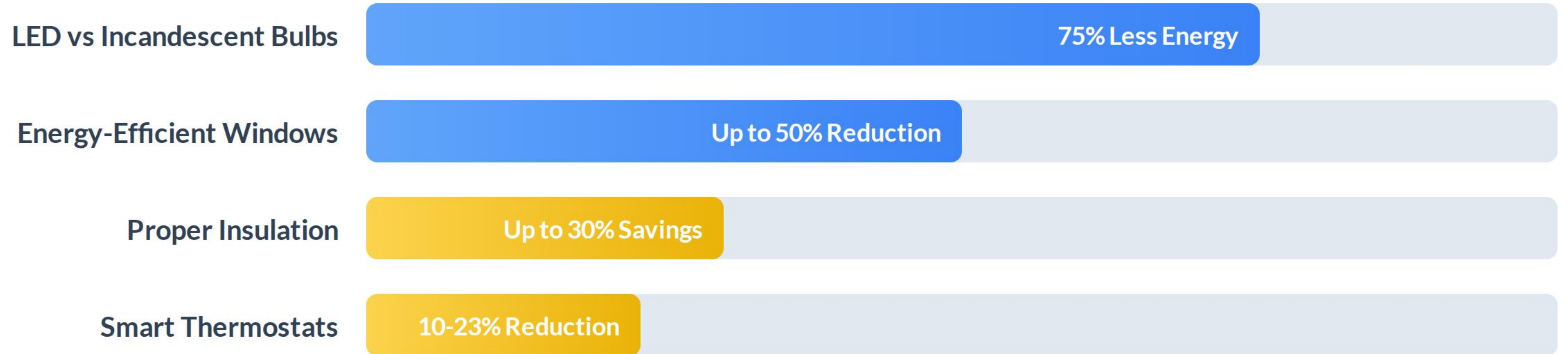
An integrated gamification feature that tests user knowledge on global energy statistics, reinforcing learning through active engagement.



Fact-Based Logic

The AI provides concrete data, such as solar panels lasting 25-30 years and reducing carbon footprints by up to 80% compared to fossil fuels.

Promoting Energy Efficiency Tactics



As demonstrated in the application's responses, energy efficiency is the fastest way to reduce consumption. The bot educates students on zero-cost and low-cost strategies to implement in hostels and campuses.

Commitment to Responsible AI

-  **Hardcoded Transparency:** The application enforces transparency. Every session begins with a mandatory greeting: *"Hi, I am an AI Mentor for Clean Energy."* Users are never misled into thinking they are speaking with a human.
-  **Financial Ethics & Safety:** To prevent risky behavior based on AI hallucinations, the bot explicitly advises users to *"consult certified experts before making financial investments"* regarding solar installations.
-  **Data Accuracy:** Responses are driven by predefined, structured logic covering established UN SDG 7 statistics to maintain high educational integrity.
-  **Inclusivity:** The mentor persona is designed to be highly encouraging and accessible, breaking down complex engineering concepts for students from all academic backgrounds.

Envisioning Real-World Impact



Empowering Campus Leaders

By deploying this Clean Energy Awareness Assistant within university networks, we transform passive students into active sustainability advocates.

Providing instant, reliable education on renewable implementation helps institutions lower their carbon footprint, ultimately bringing us closer to achieving the UN's 2030 Agenda for Sustainable Development.

Thank You!

1M1B AI for Sustainability Virtual Internship - Final Project

Questions regarding IBM BOB implementation or SDG 7 alignment?

Image Sources



Thumbnail for

Source: www.istockphoto.com

<https://media.istockphoto.com/id/492334365/photo/pollution-and-clean-energy-concept.jpg?s=612x612&w=0&k=20&c=gqqcRAWS-KioivVUXdXeZJNPP5ypbmrrh1lae3YhKFc=>



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